

a mindset shift towards building products rather than solely testing them. Continuous learning and expanding your skillset beyond just automation are vital for long-term career growth. By understanding the inner workings of systems and staying adaptable, you can carve out a successful career path beyond the boundaries of DevOps.

Q. What skills are important for DevOps roles?

A. At its core, DevOps consists of infrastructure and automation. The core skills needed to get into DevOps include a very good understanding of operating systems such as Linux, any programming language such as Python, Bash, Ruby, and PowerShell networking, and automation. These are the fundamentals.

Q. What are the tools that engineers need to learn to build those skills?

A. There are plenty of tools in DevOps, which can be overwhelming, and no one is expected to know everything. These tools and technologies can be learned as you start working with them. To get started, you can pick a few tools depending on your background and interest. Essential tools include Git for version control, Jenkins, GitLab CI or CircleCI for CI/CD, Ansible, Puppet or Chef for configuration management, Docker and Kubernetes for containerisation, Terraform or AWS CloudFormation for infrastructure as code; and Prometheus, Grafana, and the ELK Stack for monitoring and logging. Proficiency in scripting languages like Bash and Python is also important.

You don't need to learn all the above in the first place, but a strong understanding of networking, basics of operating systems, and proficiency in scripting languages like Bash and Python is crucial and important. The rest of the skills can be easily learned

once you start working on them. Tools evolve, so a strong understanding of containers and a programming language will help you adapt to new ones. Focus on mastering the basics and building a solid foundation to easily transition to new tools as needed.

Q. Are there any emerging tools or technologies that can help DevOps professionals to upskill?

A. In today's landscape, having a basic understanding of cloud computing is necessary for DevOps professionals. While it's impossible to learn everything about cloud platforms, grasping key concepts like virtual machines, virtual networks, and auto-scaling is essential. Cloud-native companies, which are prevalent nowadays, heavily rely on cloud infrastructure, making cloud skills indispensable. DevOps professionals can explore various cloud services and products, such as AWS CodePipeline for continuous integration and delivery (CI/CD) and Elastic Container Service (ECS) for container management. Transferring existing skills, like Git proficiency or Docker knowledge, to cloud platforms can be beneficial for career growth.

Schools and colleges are now focusing on cloud computing skills earlier on, which prepares students for more advanced learning and transitions into corporate environments. While specific emerging tools or technologies may vary, the overarching theme for upscaling in DevOps is mastering cloud computing concepts and leveraging cloud platforms to enhance automation and infrastructure management capabilities.

Q. How have cloud-native technologies impacted DevOps hiring?

A. Understanding cloud-native technologies has become a baseline requirement for DevOps

roles, especially for the companies established after 2010 that are built majorly on cloud infrastructure. Forums like CNCF (Cloud Native Computing Foundation) and resources provided by organisations like the Linux Foundation offer valuable insights into these technologies. CNCF has standardised frameworks around containerisation, addressing the lack of standardisation that previously existed in the field, leading to compatibility issues, as different companies built their products using different tools and frameworks. Standardisation has enabled interoperability and facilitated smoother integration of different services such as AWS CodePipeline for continuous integration and delivery (CI/CD) and Elastic Container Service (ECS) for container management.

Rather than focusing solely on vendor-specific implementations, DevOps professionals should understand the blueprint or skeleton of these technologies for more versatile skillsets that are transferable across different environments. In terms of hiring, big companies prioritise core fundamentals and the ability to explain concepts from first principles. They often have the resources and time to train individuals extensively. In contrast, startups prioritise industry-ready skillsets and value candidates who can quickly become productive.

Q. Are there some common challenges that companies face when hiring for DevOps positions?

A. One challenge many companies come across is that candidates possess in-depth knowledge of specific tools but may lack flexibility when it comes to adapting to new technologies due to deep immersion in a particular tool. This makes it challenging for any tool or framework transition as required by the company. Many individuals